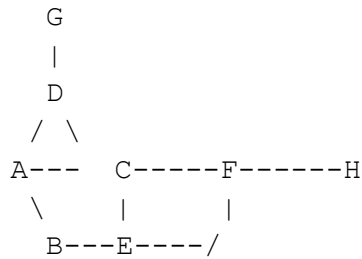


a) Social Network Diagram

Below is a worked example of an appropriate social network diagram based on the contact-tracing information provided.



Confirmed cases: A, B, C, D, E

Non-cases: F, G, H

b) Centrality Measures

High degree centrality

Individual C has the highest degree centrality, as C is directly connected to A, B, D, and F. This represents the greatest number of direct contacts in the network, increasing the potential for acquiring and transmitting infection.

High betweenness centrality

Individual F demonstrates high betweenness centrality because F lies on the shortest path between the main cluster of confirmed cases and individual H. F acts as a bridge connecting different parts of the network, and interruption at this node could reduce onward transmission.

c) Priority Targets for Public Health Intervention

Individual C should be prioritised for intervention due to high degree centrality, indicating frequent contact with multiple cases. Targeted actions could include prophylaxis, vaccination status review, exclusion from group activities, and enhanced contact follow-up.

Individual F and associated contacts (E and H) should also be prioritised. Although F is not yet a confirmed case, their bridging role poses a risk for expanding the outbreak beyond the core group. Early intervention could prevent spread to peripheral individuals.

d) Network Pattern Driving the Outbreak

The outbreak is best described as being driven by a densely connected core group with bridging individuals. Cases A through E form a tightly interconnected cluster that facilitates sustained transmission. Individual F functions as a bridge, linking the core group to otherwise peripheral individuals such as H. This structure explains continued transmission despite high population-level vaccination coverage.